

ORANGE PROBLEM ASSIGNMENT REPORT

NAME: DHARANI S

SRN: PES2UG24CS157

SECTION: C

Q7A : CRIME SCENE LASER GRID

| Code Logic

The program uses two nested loops to iterate over a 5x5 grid (rows and columns, each from 0 to 4). For every cell:

If the **row index equals the column index** (i.e., on the main diagonal), the character `#` is printed using RARS system call `ecall 11`.

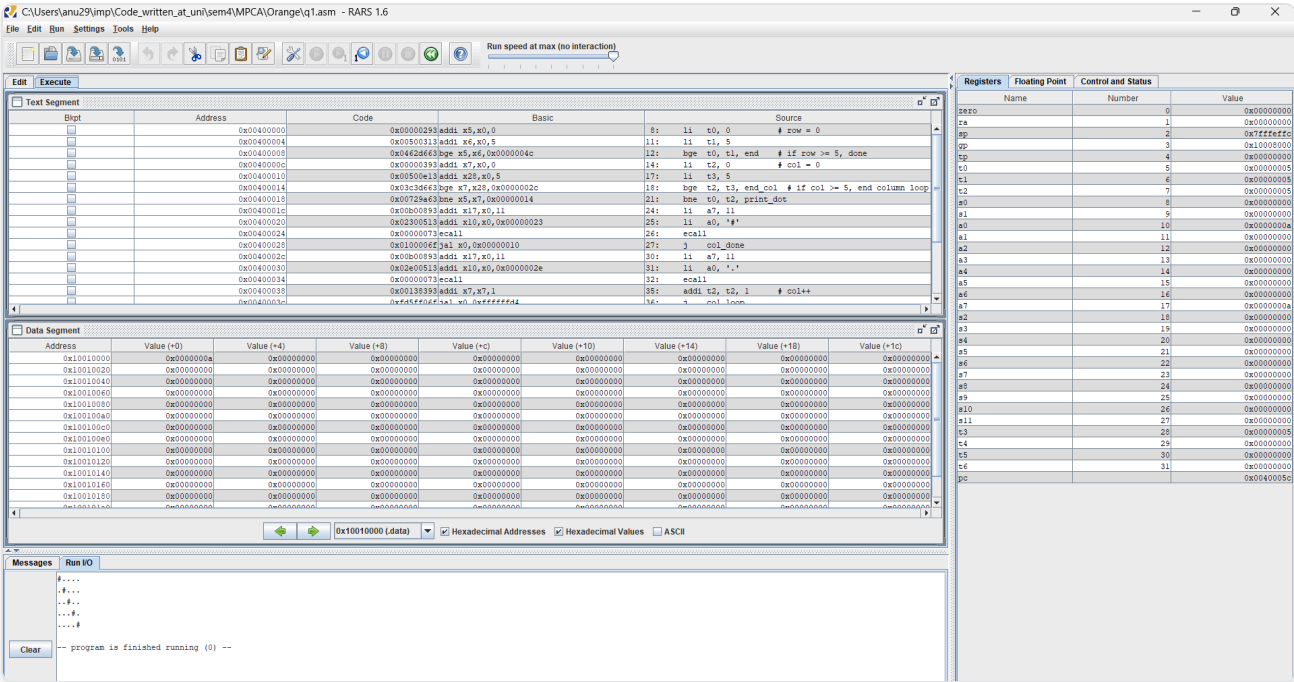
Otherwise, `.` is printed.

After each row's inner loop completes, a newline character is printed and the row counter increments. This continues until all 5 rows are processed, after which the program exits via `ecall 10`.

Registers used:

REGISTER	PURPOSE
t0	Row counter
t1	Column counter (reused as limit check temp)
t2	Column counter
t3	Temporary limit (5)
a0 , a7	System call arguments

Output Screenshot



```
Anu@windows11 ~\..\..\Orange \P main java -jar "C:\Users\anu29\Downloads\rars1_6.jar" q1.asm
RARS 1.6 Copyright 2003-2019 Pete Sanderson and Kenneth Vollmar

#...
.# ...
..#..
...#.
...#
```

Q7B: VENDING MACHINE CHANGE CALCULATOR

Code Logic

The program calculates the minimum number of notes/coins needed to return change to a customer.

Given values:

Price = 65

Amount Paid = 100

Change = $100 - 65 = 35$

The denominations used (in descending order): 50, 20, 10, 5, 1

For each denomination, the program:

Divides the remaining change by the denomination using `div` → gets the count.

Takes the remainder using `rem` → updates remaining change.

Prints the denomination label and count using `ecall 4` (string) and `ecall 1` (integer).

This greedy approach ensures the minimum number of notes/coins is always used.

Registers used:

REGISTER

PURPOSE

t0

Price

t1

Amount paid

t2

Remaining change

t3

Current denomination

t4

Count of that denomination

a0

, a7

System call arguments

Output:

C:\Users\anu29\mp\Code_written_at_uni\sem4\MPCA\Orange\q2.asm - RARS 1.6

File Edit Run Settings Tools Help

Run speed at max (no interaction)

Text Segment

Bitpt	Address	Code	Basic	Source
	0x00400000	0x04100293addi x5,x0,0x00000041	14:	li t0, 65 # price
	0x00400004	0x04400313addi x6,x0,0x00000044	15:	li t1, 100 # amount paid
	0x00400008	0x040303b3sub x7,x6,x5	16:	sub t2, t1, t0 # change = paid - price => ...
	0x0040000c	0x00400093addi x17,x0,4	19:	li a7, 4
	0x00400010	0x0fc10517auipc x10,0x00000fc10	20:	la a0, msg_change
	0x00400014	0x0ff05013addi x10,x10,0xfffffff0		
	0x00400018	0x00000007jccall	21:	ecall
	0x0040001c	0x00100093addi x17,x0,1	24:	li a7, 1
	0x00400020	0x00700533addi x10,x0,x7	25:	mv a0, t2
	0x00400024	0x00000007jccall	26:	ecall
	0x00400028	0x00400093addi x17,x0,4	29:	li a7, 4
	0x0040002c	0x0fc10517auipc x10,0x00000fc10	30:	la a0, newline
	0x00400030	0x00550513addi x10,x10,5		
	0x00400034	0x00000007jccall	31:	ecall
	0x00400038	0x03200e13addi x28,x0,0x00000032	34:	li t3, 50
	0x0040003c	0x0b3c0c03div x26,x7,x28	35:	div r4, r3, r3 # count of 50s

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010064	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010068	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001006c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010070	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010074	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010078	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001007c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010084	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010088	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001008c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010090	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010094	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010098	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001009c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a4	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a8	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100ac	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100b0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers

Name	Number	Value
\$zero	0	0x00000000
\$a0	1	0x00000000
\$a1	2	0x7fffffff
\$a2	3	0x10000000
\$a3	4	0x00000000
\$a4	5	0x00000041
\$a5	6	0x00000044
\$a6	7	0x00000000
\$a7	8	0x00000000
\$s0	9	0x00000000
\$s1	10	0x10010031
\$s2	11	0x00000000
\$s3	12	0x00000000
\$s4	13	0x00000000
\$s5	14	0x00000000
\$s6	15	0x00000000
\$s7	16	0x00000000
\$s8	17	0x00000004
\$s9	18	0x00000000
\$t0	19	0x00000000
\$t1	20	0x00000000
\$t2	21	0x00000000
\$t3	22	0x00000000
\$t4	23	0x00000000
\$t5	24	0x00000000
\$t6	25	0x00000000
\$t7	26	0x00000000
\$t8	27	0x00000000
\$t9	28	0x00000000
\$t10	29	0x00000000
\$t11	30	0x00000000
\$t12	31	0x00000000
\$pc		0x0040014c

Messages

Run IO

program is finished running (0) --

Change: 35

50s: 0

20s: 1

10s: 1

5s: 1

1s: 0

program is finished running (0) --

```

Anu@windows11 ~\..\..\Orange % main java -jar "C:\Users\anu29\Downloads\rars1_6.jar" q2.asm
RARS 1.6 Copyright 2003-2019 Pete Sanderson and Kenneth Vollmar

Change: 35
50s: 0
20s: 1
10s: 1
5s: 1
1s: 0

```